

NIVALA, Finland

WMO: **029050**

Lat: **63.917N** Lon: **24.967E** Elev: **81** StdP: **100.36** Time Zone: **2.00 (EUE)** Period: **90-00** WBAN: **99999**

Annual Heating, Humidification, and Ventilation Design Conditions

	Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF	
				99.6%			99%			0.4%		1%					
		99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(1)	2	-26.1	-22.6	-28.5	0.3	-26.0	-24.8	0.4	-22.5	12.0	3.0	10.2	2.6	1.3	160	0.667	(1)

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

	Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB		
			0.4%		1%		2%		0.4%		1%		2%				
			DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(2)	7	9.2	25.4	17.4	23.5	16.4	21.8	15.6	18.5	22.8	17.4	22.0	16.5	20.6	3.2	150	(2)

	Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB	
	0.4%			1%			2%			0.4%		1%		2%			
	DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(3)	16.9	12.1	20.7	15.6	11.2	19.0	14.6	10.5	18.1	52.5	22.5	49.3	22.2	46.5	20.5	22.2	(3)

Extreme Annual Design Conditions

	Extreme Annual WS				Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
					Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
	1%	2.5%	5%		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
	(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4)	9.2	7.9	6.9	DB	-29.9	28.1	3.5	1.9	-32.4	29.5	-34.4	30.6	-36.3	31.6	-38.8	33.0	(4)
(5)				WB	-30.0	20.4	3.4	1.4	-32.5	21.4	-34.4	22.2	-36.4	23.0	-38.8	24.0	(5)

Monthly Climatic Design Conditions

				Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
				(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg		2.9	-7.5	-8.1	-3.4	1.5	7.2	13.3	15.5	13.5	8.0	2.5	-3.3	-5.7	(6)
(7)		DBStd		9.58	6.45	7.53	4.75	4.48	4.16	3.68	2.81	3.25	3.68	4.66	5.30	6.44	(7)
(8)		HDD10.0		3054	544	507	416	257	109	11	1	7	83	232	400	487	(8)
(9)		HDD18.3		5667	802	740	674	506	345	154	96	153	311	490	650	746	(9)
(10)		CDD10.0		444	0	0	0	1	23	111	171	115	22	1	0	0	(10)
(11)		CDD18.3		16	0	0	0	0	0	5	8	3	0	0	0	0	(11)
(12)		CDH23.3		171	0	0	0	0	7	57	76	31	0	0	0	0	(12)
(13)		CDH26.7		19	0	0	0	0	0	6	12	1	0	0	0	0	(13)
(14)	Wind	WSAvg		3.0	3.2	3.1	3.3	3.0	3.0	2.9	2.8	2.6	2.8	3.3	2.9	3.0	(14)
(15)	Precipitation	PrecAvg		602	40	32	31	32	46	63	79	71	54	57	50	47	(15)
(16)		PrecMax		771	77	64	51	70	87	115	136	143	124	106	97	98	(16)
(17)		PrecMin		456	11	4	4	8	7	21	20	27	19	13	5	17	(17)
(18)		PrecStd		82	18	16	12	15	22	27	27	30	31	25	21	20	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB		4.6	5.7	8.2	17.1	24.3	27.4	28.5	25.8	20.4	12.9	7.3	4.1	(19)
(20)			MCWB		2.7	2.9	4.8	10.5	15.3	18.4	20.0	17.8	15.6	11.0	6.3	3.0	(20)
(21)		2%	DB		2.6	3.7	5.5	13.7	21.5	25.1	25.5	24.0	17.6	11.0	5.7	3.1	(21)
(22)			MCWB		1.6	2.3	2.7	8.3	13.5	16.9	18.0	17.3	13.7	9.6	4.8	2.3	(22)
(23)		5%	DB		1.5	2.2	4.0	11.2	18.3	22.7	23.4	22.2	15.4	9.8	4.3	2.3	(23)
(24)			MCWB		0.7	1.2	1.7	6.8	11.6	15.6	16.7	16.5	12.6	8.5	3.6	1.6	(24)
(25)		10%	DB		0.5	1.2	2.6	8.5	15.5	20.7	21.5	20.2	14.0	8.5	2.8	1.4	(25)
(26)			MCWB		-0.1	0.4	1.1	5.0	10.1	14.7	15.8	15.4	11.9	7.4	2.1	0.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB		2.9	3.9	5.3	11.2	15.5	20.0	21.0	18.9	16.1	11.4	6.7	3.3	(27)
(28)			MCDB		4.2	5.0	7.7	16.3	21.6	26.0	27.3	22.9	19.2	12.6	7.2	3.7	(28)
(29)		2%	WB		1.8	2.5	3.2	8.6	14.0	17.9	19.0	17.8	14.5	9.8	4.9	2.4	(29)
(30)			MCDB		2.5	3.5	4.9	13.3	21.1	22.6	23.0	22.4	16.7	10.8	5.5	2.9	(30)
(31)		5%	WB		0.9	1.5	2.1	7.0	12.3	16.6	17.6	17.0	13.2	8.7	3.7	1.7	(31)
(32)			MCDB		1.4	2.2	3.5	10.6	17.4	21.4	21.9	21.6	15.0	9.6	4.3	2.2	(32)
(33)		10%	WB		-0.1	0.5	1.3	5.4	10.5	15.2	16.7	15.8	11.8	7.4	2.1	0.8	(33)
(34)			MCDB		0.5	1.2	2.4	8.2	14.8	20.0	20.5	19.6	13.5	8.3	2.7	1.2	(34)
(35)	Mean Daily Temperature Range		MDBR		6.6	6.9	8.1	8.9	10.4	9.6	9.2	9.3	8.0	5.1	4.7	5.7	(35)
(36)		5% DB	MCDBR		5.5	5.9	8.4	13.0	15.6	13.2	13.1	12.9	9.8	5.7	4.8	4.7	(36)
(37)			MCWBR		5.3	5.3	6.2	8.2	8.6	6.7	6.5	6.8	6.4	4.5	4.4	4.5	(37)
(38)		5% WB	MCDBR		5.3	5.6	6.7	12.3	14.1	11.5	11.1	11.6	8.3	5.3	4.9	4.8	(38)
(39)			MCWBR		5.2	5.3	5.1	7.8	8.1	6.4	6.0	6.5	5.8	4.7	4.6	4.7	(39)
(40)	Clear-Sky Solar Irradiance	taub			0.218	0.282	0.310	0.351	0.346	0.348	0.369	0.353	0.329	0.308	0.361	0.356	(40)
(41)		taud			1.987	2.136	2.196	2.317	2.443	2.457	2.416	2.456	2.512	2.378	2.637	2.631	(41)
(42)		Ebn at Noon			357	624	771	824	867	871	837	819	768	608	331	182	(42)
(43)		Edh at Noon			28	62	89	100	97	97	99	87	67	50	25	16	(43)
(44)	All-Sky Solar Radiation	RadAvg			0.15	0.70	2.18	3.74	4.88	5.37	5.01	3.79	2.15	0.81	0.19	0.06	(44)
(45)		RadStd			0.02	0.13	0.21	0.36	0.53	0.47	0.50	0.39	0.23	0.15	0.04	0.01	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (43 neighbors)	+0.51	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page